

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX)		**Date:31-12-24**
Time: 3HRS	**GTM-7**	**Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. A concave lens of glass of refractive index 1.5, has both surfaces of same radius of curvature R. On immersion in a medium of refractive index 1.75, it will behave as a:

   A) convergent lens of focal length 3.5 R

   B) convergent lens of focal length 3.0 R

   C) divergent lens of focal length 3.5 R

   D) divergent lens of focal length 3.0 R

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2. In a photo emissive cell with exciting wavelength λ, the fastest electron has a speed V. If the exciting wavelength is changed to $\frac{3\lambda}{4}$, the speed of the fastest emitted electrons will be

A) $V\left(\frac{4}{3}\right)^{1/2}$ B) $V\left(\frac{3}{4}\right)^{1/2}$

C) less than $V\left(\frac{4}{3}\right)^{1/2}$ D) greater than $V\left(\frac{4}{3}\right)^{1/2}$

3. A hydrogen atom and an Li^{++} ion are both in the second excited state. If l_H and l_{Li} are their respective electronic angular momentum , E_H and E_{Li} are their respective energies, then

A) $l_H > l_{Li}$ and $|E_H| > |E_{Li}|$ B) $l_H = l_{Li}$ and $|E_H| < |E_{Li}|$

C) $l_H = l_{Li}$ and $|E_H| > |E_{Li}|$ D) $l_H < l_{Li}$ and $|E_H| < |E_{Li}|$

4. The scale of a galvanometer is divided into 150 equal divisions. The galvanometer has current sensitivity of 10 divisions per mA and a voltage sensitivity of 2 divisions per mV. Find the shunt resistance so that it can read 6A per division.

A) $6.2\times10^{-5}\Omega$ B) $3.6\times10^{-5}\Omega$ C) $9.2\times10^{-5}\Omega$ D) $8.3\times10^{-5}\Omega$

5. Under the influence of the Coulomb field of a fixed charge $+Q$ at point O, a charge $-q$ is moving around it in an elliptical orbit. Find out the correct statement

A) The angular momentum of the charge $-q$ is conserved about the point O

B) The linear momentum of the charge $-q$ is conserved about the point O

C) The angular velocity of the charge $-q$ is conserved about the point O

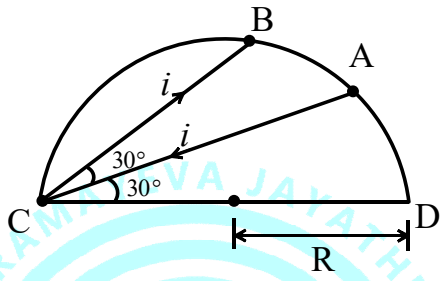
D) The linear speed of the charge $-q$ is conserved about the point O

6. If an ideal gas is at a temperature $T\,K$ and pressure P and the diameter of its molecule is d, and the rms speed of the molecules is v, then the average time available in between two successive collisions is $(k_B$:Boltzmann's constant$)$

A) $\frac{k_BT}{\sqrt{2}\pi Pvd^2}$ B) $\frac{k_BT}{\sqrt{2}Pvd^2}$ C) $\frac{k_BT}{\pi Pvd^2}$ D) $\frac{k_BT}{2\pi Pvd^2}$

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7. A current carrying wire is placed in the grooves of an insulated semi circular disc of radius 'R', as shown. The current enters at point A and leaves from point B. Determine the magnetic field at point D.

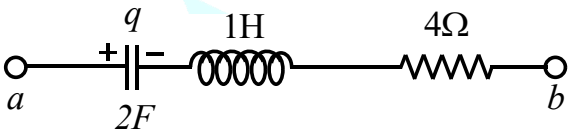


A) $\frac{\mu_0 I}{8\pi R\sqrt{3}}$ B) $\frac{\mu_0 I}{4\pi R\sqrt{3}}$ C) $\frac{\mu_0 I}{4\pi R}$ D) None of these

8. Speed of a particle moving in a circular path of radius $r = 2m$ varies with time t as $v = t^2$, where t is in second and v in m/s. Find the net acceleration at $t = 2s$.

A) $\sqrt{80}\, m/s^2$ B) $\sqrt{40}\, m/s^2$ C) $\sqrt{20}\, m/s^2$ D) $\sqrt{34}\, m/s^2$

9. In the circuit shown the charge (in C) on the capacitor varies with time (in s) as $q = \left(t^2 - 16\right)$. Find $V_a - V_b$ at $t = 5s$.



A) $50V$ B) $35.5V$ C) $46.5\,V$ D) $40.2\,V$

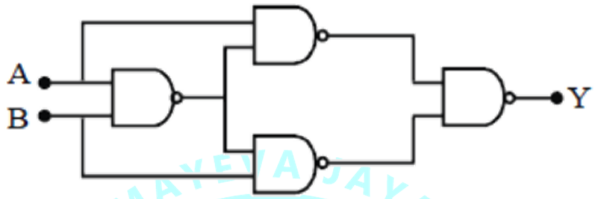
10. The units of $\frac{CV}{\rho\varepsilon_0}$ is the same as that of

(C :capacitance; V : Voltage; ρ : Specific resistance; ε_0: Absolute permittivity)

A) Charge B)Current C) Time D) Frequency

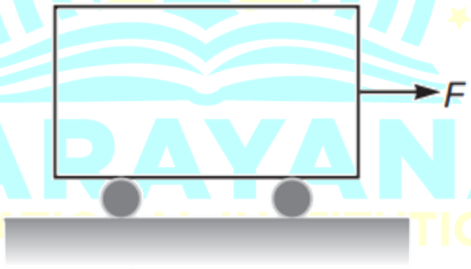
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11. Select the outputs Y of the combination of gates shown below for inputs A =1, B = 0; A =1, B=1and A =0,B = 0 respectively



A) (0 1 1) B) (0 0 1) C) (1 0 0) D) (1 1 1)

12. A car of mass m is accelerating on a level smooth road under the action of a single force F. The power delivered to the car is constant and equal to P. If the velocity of the car at an instant is v, then find the distance after which the velocity gets doubled.



A) $\frac{7mv^3}{3P}$ B) $\frac{4mv^3}{3P}$ C) $\frac{mv^3}{P}$ D) $\frac{18mv^3}{7P}$

13. Match List-1 with List-II

List-I	**List-II**
EM-Wave	**wave length range**
1) gamma rays	(I) about$10^{-3}m$ *to* $10^{-1}m$
2) visible light	(II) about$10^{-6}m$ *to* $10^{-3}m$
3) infrared rays	(III) about$10^{-7}m$
4) micro waves	(IV) *less than* $10^{-12}m$

A) $1-IV, 2-III, 3-I, 4-II$ B) $1-IV, 2-III, 3-II, 4-I$

C) $1-III, 2-IV, 3-I, 4-II$ D) $1-III\ 2-II, 3-IV, 4-I$

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14. If the ratio $C_p / C_v = \gamma$, the change in internal energy of the given gas when the volume changes from V to 2V at constant pressure P is

A) $\frac{R}{(\gamma-1)}$ B) PV C) $\frac{PV}{(\gamma-1)}$ D) $\frac{\gamma PV}{(\gamma-1)}$

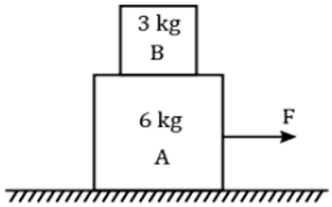
15. If the distances of an object and its virtual image from the focus (on the same side as the image and object) of a convex lens are 1 cm each, then the focal length of the lens $(in\ cm)$ is:

A) 4 B) $\sqrt{2}+1$ C) $2\sqrt{2}$ D) $2+\sqrt{2}$

16. The time taken by a particle performing SHM to pass from point A to B where its velocities are same is $2\,s$. After another $2s$ it returns to B. The time period of oscillation is (in seconds)

A) 2 B) 8 C) 6 D) 4

17. Two blocks A and B of masses 6 kg and 3 kg rest on a smooth horizontal surface as shown in adjoining Fig. If coefficient of friction between A and B is 0.4, the maximum horizontal force which can make them move without B sliding on A is



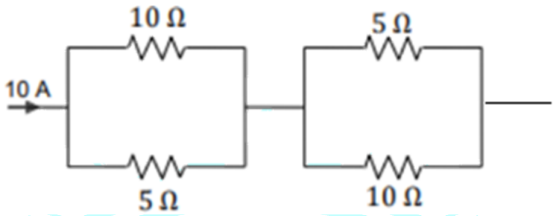
A) 72 N B) 40N C) 36N D) 20 N

18. Orbital period of a planet is 8 years if the semi major axis of its orbit $4AU$. The orbital period of another planet with semi major axis of $9AU$ is

A) 12 years B) 16 years C) 20 years D) 27 years

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19. Four resistances carrying a current shown in figure are immersed in a box containing ice at 0°C. How much ice must be put in the box every 10 min to keep the average quantity of ice in the box constant? Latent heat of ice is 80 cal g^{-1}



A) 1.190 kg B) 3.20 kg C) 4.2 kg D) 0.25 kg

20. The ratio of radii of nuclei $_{13}Al^{27}$ and $_{52}X^{A}$ is $3:5$. The number of neutrons in the nuclei of X will be

A) 52 B) 73 C) 125 D) 13

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

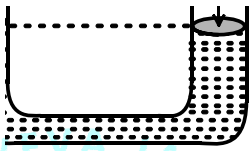
This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. An ideal solenoid of length 0.4 m and having 500 turns of wire carries a current of $3A$. The torque required to hold a coil (having radius 0.02 cm, current $2A$ and turns 50) in the middle of the solenoid with its axis perpendicular to the axis of the solenoid is $\frac{3}{x}\times10^{-7}\ Nm$ (Take $\pi^2 = 10$). Find is the value of x.

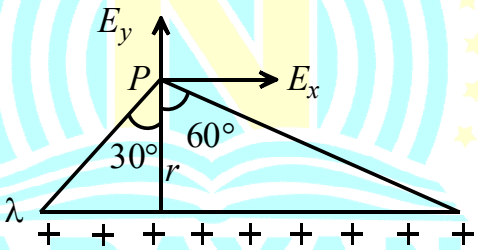
22. A Solid sphere is rolling without slipping a plane surface about its axis of symmetry. The ratio of translational kinetic energy to its total kinetic energy is $\frac{x}{7}$. The value of x is___

23. The area of cross-section of the two arms of a hydraulic press is 1 cm^2 and 20 cm^2 respectively as shown in given Fig. A force of 10 N is applied on the water in the thinner arm. What force should be applied on the water in the thicker arms so that the water may remain in equilibrium? (in N)

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24. Resultant the electric field at point P due to the given wire is $E = \frac{k\lambda}{2r}\sqrt{x}$. The value of x is



25. A piece of conducting wire of resistance R is cut into $2n$ equal parts. Half the parts are connected in series to form a bundle and remaining half in parallel to form another bundle. These bundles are then connected to give the maximum resistance. The maximum resistance of the combination is $\frac{R}{2}\left(N + \frac{1}{n^2}\right)$, the value of N is

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Match List – I with List – II

	List – I		**List – II (L.P on central atom)**
1)	ClF_3	(i)	1
2)	IF_5	(ii)	2
3)	ClF	(iii)	4
4)	CCl_4	(iv)	3
		(v)	0

Choose the correct answers

A) $1-ii, 2-i, 3-iv, 4-iii$ B) $1-ii, 2-iii, 3-v, 4-i$

C) $1-i, 2-v, 3-iv, 4-ii$ D) $1-ii, 2-i, 3-iv, 4-v$

27. Given below are two statements; one is labelled as Assertion (A) and other is labelled as Reason (R)

Assertion (A) : H_2O and H_2S molecules have bent shape and resultant dipole moment of H_2O is less than H_2S

Reason (R): 'Sulphur' has less electro negativity than 'oxygen'.

In the light of the above statements choose the correct answer from the options given below

A) Both A & R are true and (R) is the correct explanation of (A)

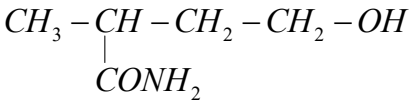
B) (A) is false but (R) is true

C) (A) is true but (R) is false

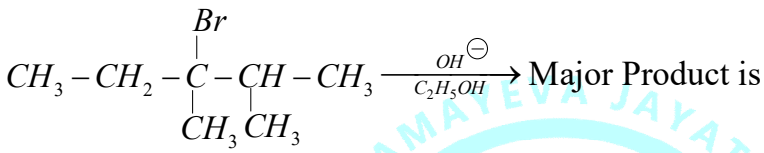
D) Both (A) and (R) true but (R) is not correct explanation of A

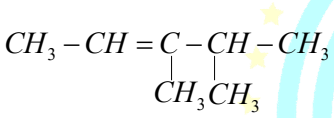
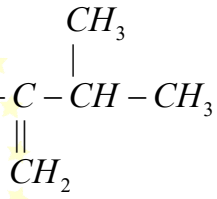
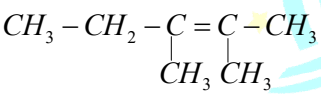
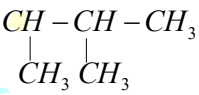
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28.  Correct IUPAC name of the compound is

A) 3-Amido-1-butanol
B) 4-Hydroxy-2-methylbutanamide
C) 3-Methyl-1-hydroxy-butanamide
D) 1-Oxo-1-amino-2-methylbutane-1-ol

29.  Major Product is

A) 
B) 
C) 
D) 

30. Calculate the amount of copper deposited when 20amp of current is passed for 9650 sec through $CuSO_4$ solution (Cu atomic wt = 63.5 g/mol)

A) 1 Chemical equivalent of Cu
B) 1 gram atom of Copper
C) 0.1 gram atoms of copper
D) 31.75g of Cu

31. The weight of ethane required to produce 2.27 lit of CO_2 at STP(273k, 1bar). on combustion

A) 30g
B) 0.3g
C) 1.5g
D) 3g

32. $E^0_{Ag^+/Ag} = 0.8V, E^0_{Mg^{+2}/Mg} = -2.36V$. When these two electrodes are used to construct electro chemical (voltaic) Cell.

A) $2Ag_{(s)} + Mg^{+2}_{(aq)} \rightarrow 2Ag^{+}_{(aq)} + Mg_{(s)}$ Reaction is spontaneous

B) Ag^+ undergoes reduction

C) Mg^{+2} undergoes reduction

D) Ag undergoes reduction

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33. In the precipitation of the iron group in qualitative analysis ammonium chloride is added before adding ammonium hydroxide to

A) Decrease concentration of OH^- ions B) Prevent interference by phosphate ions

C) Increase concentration of NH_4^+ ions D) Increase concentration of Cl^- ions

34. Given below are two statements.

Satement-1: Atomic radius of 2nd period elements decreases in a period from left to right upto halogens

Statement-2: Oxygen atomic radius is greater than fluorine and fluorine atomic radius is greater than Neon

In the light of the above statements choose the correct answer from the options given below.

A) Statement -1 is correct, statement -2 is false

B) Statement -1 and statement -2 both are correct

C) Statement -1 and statement -2 both are false

D) Statement -1 is false statement -2 is correct

35. Which of the following is most reactive towards aqueous HBr

A) 1-phenyl-2-propanol B) 1-phenyl-1-propanol

C) 3-phenyl-1-propanol D) 2-phenyl-1-propanol

36. Which of the following statements is **FALSE**?

A) In an octahedral crystal field the d-electrons in a metal ion occupy the e_g set before occupying the t_{2g} set of orbitals

B) Diamagnetic metal ions cannot have an odd number of electrons

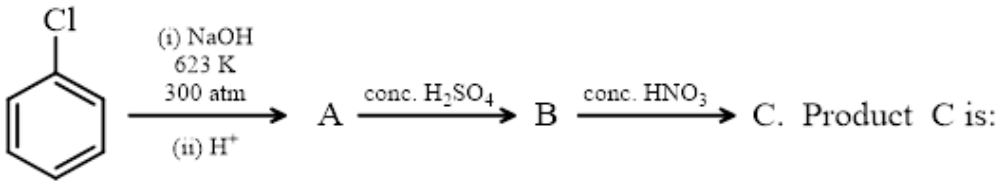
C) Low spin complexes can be paramagnetic

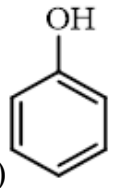
D) In high spin octahedral complexes Δ_0 is less than the pairing energy and is relatively small

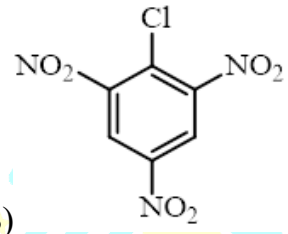
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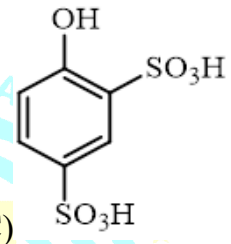
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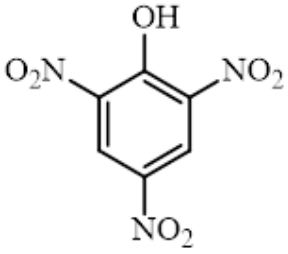
37.



A) 

B) 

C) 

D) 

38. The correct statements from the following are

a) In 13th group elements 'B' has highest melting point

b) +3 oxidation state does not exist for In

c) 'B' cannot liberate H_2 gas with Acids

d) Electronegativity order is $B > Tl > In > Ga > Al$

Choose the correct answer from the options given below

A) a, b, c, d B) a, b only C) a, c, d D) b, c only

39. Which complex is likely to show optical activity?

A) $Trans\left[Ma_4b_2\right]^{\pm n}$ B) $\left[Ma_5b\right]^{\pm n}$

C) $cis\left[Ma_2\left(AA\right)_2\right]^{\pm n}$ D) $Trans\left[Ma_2\left(AA\right)_2\right]^{\pm n}$

40. Set of essential amino acids are

A) Lysine, threonine, Tyrosine B) Methionine, Glutamine, Arginine

C) Leucine, Valine, Alanine D) Histidine, Phenylalanine, Lysine

41. Hept-4-ene-2-ol on treatment with PCC gives 'A'. 'A' on reaction with sodium hypoiodite gives 'B' and an yellow solid. Which on further heating with soda lime gives 'C'. The compound 'C' is.

A) 2-Butene B) 2-Pentene C) Propanaldehyde D)4-methylpent-2-ene

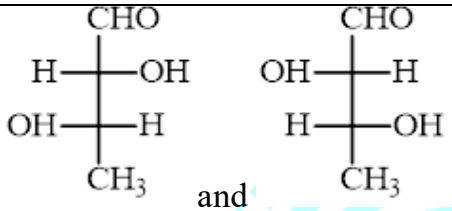
42. The number of ions from the following that cannot liberate hydrogen from dilute HCl solution

(i) Ti^{+2} (ii) V^{+2} (iii) Cu^{+2} (iv) Cr^{+2}

A) 1 B) 0 C) 3 D) 2

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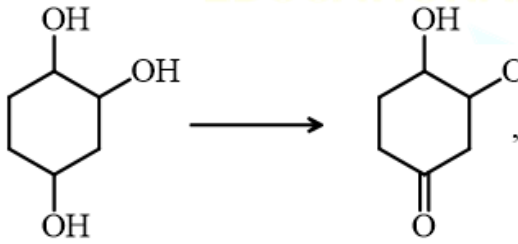
43. Match the columns

	Column – I		Column – II
a)	CH_3COOH and $HCOOCH_3$	(i)	Functional
b)	 and	(ii)	Position
c)	1-Butene & 2 – Butene	(iii)	Metamers
d)	Diethyl ether & Methyl Propyl ether	(iv)	Enantiomers

Correct Match

A) $a-ii, b-i, c-iii, d-iv$ B) $a-i, b-iv, c-ii, d-iii$

C) $a-iii, b-iv, c-ii, d-i$ D) $a-iv, b-ii, c-iii, d-i$

44.  , This conversion can be achieved by:

A) $Me_2CO / H^+, H_3O^{\oplus}, KMnO_4 / HO^-$ B) $Me_2CO / H^+, KMnO_4, H_3O^+$

C) $KMnO_4 / NaO_4, Me_2CO / H^+, H_3O^+$ D) $KMnO_4 / NaIO_4, H_3O^+, Me_2CO / H^+$

45. **Statement – I:** Na_2SO_4 is soluble in water but $BaSO_4$ is insoluble

Statement – II : In case of $BaSO_4$ its lattice energy is more than its hydration energy.

In the light of the above statements choose the most appropriate answer from the options given below.

A) Statement – 1 & Statement – II are correct

B) Statement – I is correct but statement – II is wrong

C) Statement – I is wrong but statement – II is correct

D) Statement – I & Statement – II both are wrong

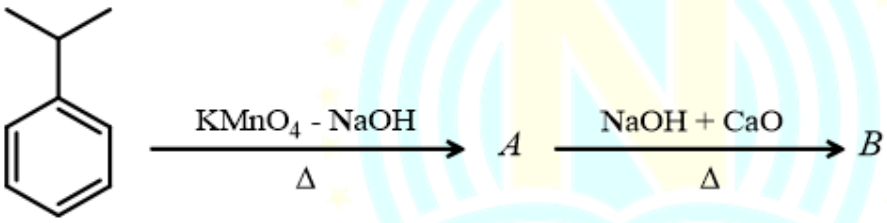
APEX JEE-MAIN 2025
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SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Maximum number of electrons associated with a principal Quantum number $n=4, l=1, m=\pm 1$

47.



In the product 'B' the ratio of σ and π bonds is?

48. Calculate the magnitude of enthalpy change for the reaction(in kJ/mole)

$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)}$.

The enthalpy of formation of $CH_{4(g)}, CO_{2(g)}$ and $H_2O_{(l)}$ are -75 kJ/mole, -393.5kJ/mole and -285.8 kJ/mole respectively

49. For the reaction $2A + B \rightarrow A_2B$, rate $= k[A][B]^2$ with $k = 2.0 \times 10^{-6} mol^{-2} L^2 s^{-1}$ the initial rate of the reaction when $[A] = 0.1$ mol/lit, $[B] = 0.2$ mol/lit is $x \times 10^{-9} mol\, L^{-1} s^{-1}$; the rate of reaction after $[A]$ is reduced to 0.06 mol/lit is $y \times 10^{-9} mol\, L^{-1} s^{-1}$ then $x + y$ is ____ (nearest integer)

50. The vapour pressure of a liquid decreases by 10 torr when a non volatile solute is dissolved, the mole fraction of the solute in solution is 0.1. Mole fraction of the liquid would be $a \times 10^{-1}$ if the decrease in vapour pressure is 20 torr, the same solute being dissolved then 'a' is____

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MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let $f_n:(-2,1)\to R$ be defined by $f_n(x)=\left|[x+n]\right|+\sqrt{x-[x]}$ and a_n denotes number of points of discontinuity of the function $f_n(x)$. Then $\sum_{r=0}^{2025} a_r$ is equal to (where $[\bullet]$ denotes GIF)

A) 4 B) 3 C) 4052 D) 2027

52. The differential equation of the family of circles touching x-axis at the origin and center on y-axis is equal to

A) $(x^2+y^2)dy=2xy\,dx$ B) $(x^2+y^2)dy=4xy\,dx$

C) $(x^2-y^2)dy=2xy\,dx$ D) $(x^2-y^2)dy=4xy\,dx$

53. The area of the region enclosed by the locus of 'z' satisfying $\log_{\sqrt{3}/2}\left(\frac{|z-2|+5}{4|z-2|-4}\right)<2$ is

A) 16π B) 30π C) 25π D) 15π

54. A water jet from a fountain reaches its maximum height of 4m at a distance 0.5 m from the vertical passing through the point O of water outlet. The height of the jet above the horizontal OX at a distance of 0.75 m from the point O is (consider path of waterjet as parabola)

A) 5m B) 6m C) 3m D) 7m

55. 720 words can be made using all the letters of the word 'ACNIHS' with or without meaning. If these words are written as in a dictionary, then the 601th word is

A) SHINCA B) SNICHA C) NISAHC D) SACHIN

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56. Let d_n be the distance travelled by a fighter plane in the nth hour. For example: d_1 is the distance travelled during first hour, d_2 is the distance travelled during the second hour and so on. If the distances $d_1, d_2, d_3, ..., d_n, d_{n+1}$ satisfy the relation $d_{n+1} = 2d_n + n\left(1 + 2^n\right), d_1 = 1$ where $n \le 10$ .If for some k, $d_k = (k^2 - 2k + 13)2^{k-2} - k - 1$ then the value of k is equal to:

A) 6 B) 7 C) 8 D) 9

57. Let $\vec{r} = (\vec{a} \times \vec{b})\sin x + (\vec{b} \times \vec{c})\cos y + 2(\vec{c} \times \vec{a})$ where $\vec{a}, \vec{b}, \vec{c}$ are non-zero and non-coplanar vectors. If $\vec{r}$ is orthogonal to $\vec{a} + \vec{b} + \vec{c}$, then minimum value of $\frac{20}{\pi^2}(x^2 + y^2) =$

A) 20 B) 25 C) 30 D) 35

58. The number of integral values of 'b' for which the origin and the point (1, 1) lie on the same side of straight line $a^2x + aby + 1 = 0$ for $a \in R - \{0\}$ is

A) 3 B) 2 C) 5 D) 1

59. A piece of cheese is located at (12, 10) in a coordinate plane. A mouse is at $(4, -2)$ and is running up the line $y = -5x + 18$. At the point (h, k) the mouse starts getting farther from the cheese rather than closer to it. The value of $h + k$ is

A) 6 B) 2 C) 8 D) 10

60. The greatest value of the term independent of x in the expansion of $\left(x\sin\alpha + \frac{\cos\alpha}{x}\right)^{10}$ is

A) 2^5 B) ${}^{10}C_5$ C) $\frac{1}{32} \cdot {}^{10}C_5$ D) $\frac{1}{16} \cdot {}^{10}C_5$

61. If the domain of the function $f(x) = \frac{\sqrt{x^2 - 25}}{\left(4 - x^2\right)} + \log_{10}\left(x^2 + 2x - 15\right)$ is $(-\infty, \alpha) \cup [\beta, \infty]$, then $\alpha^2 + \beta^3$ is equal to

A) 140 B) 150 C) 125 D) 175

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62. The number of integral values of parameter a for which three distinct chords of the ellipse $\frac{x^2}{2a^2}+\frac{y^2}{a^2}=1$ (other than its diameter) passing through the point $P\left(11a,-\frac{a^2}{4}\right)$ are bisected by the parabola $y^2=4ax$ is

A) 5 B) 2 C) 3 D) 0

63. Let the point $(3,h,k)$ lie on the line of shortest distance between the lines $\frac{x+2}{-1}=\frac{y+6}{2}=\frac{z-1}{0}$ and $\frac{x+2}{-3}=\frac{y-2}{4}=\frac{z-5}{2}$. Then $h+k$ = _____

A) 3 B) – 3 C) 2 D) – 2

64. If the system of linear equations

$$x+ky+3z=0,$$
$$3x+ky-2z=0,$$
$$2x+4y-3z=0$$

has a non-zero solution (x,y,z) then $\frac{xz}{y^2}$ =

A) – 10 B) 30 C) 10 D) – 30

65. A quadratic equation is chosen from the set of all the quadratic equations which are unchanged by squaring their roots. The probability that chosen equation has equal roots is

A) $\frac{1}{3}$ B) $\frac{1}{2}$ C) $\frac{1}{4}$ D) $\frac{2}{3}$

66. The circle S touches the sides AB and AD of the rectangle ABCD and cuts the side DC at single point F and the side BC at a single point E. If $|AB|=32$, $|AD|=40$ and $|BE|=1$, then radius of circle is

A) 22 B) 25 C) 41 D) 27

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67. Let $\alpha \in (0,1)$ and $\beta = \log_e(1-\alpha)$. Let $P_n(x) = x + \frac{x^2}{2} + \frac{x^3}{3} + ... + \frac{x^n}{n}$, $x \in (0,1)$. Then $\int_0^{\alpha} \frac{t^{50}}{1-t} dt =$

A) $\beta - P_{50}(\alpha)$ B) $P_{50}(\alpha) - \beta$ C) $-\beta - P_{50}(\alpha)$ D) $\beta + P_{50}(\alpha)$

68. Let $A = (a_{ij})$ be 3×3 matrix defined as $a_{ij} = \begin{cases} i-j, & i \neq j \\ i^2 - j^2, & i = j \end{cases}$. Then $\det(\text{adj}\, A) =$

A) 25 B) 16 C) 0 D) 64

69. If $x = \sec\theta - \cos\theta$ and $y = \sec^5\theta - \cos^5\theta$ then $(x^2+4)(y')^2 = k^2(y^2+4)$, then $k =$ ____ $\left(y' = \frac{dy}{dx}\right)$

A) 5 B) 25 C) $\frac{1}{25}$ C) 1

70. For $x \in (1,\infty)$, if $P = x^3 - \frac{1}{x^3}$ and $Q = x - \frac{1}{x}$ then minimum value of $\frac{P}{Q^2} =$

A) $4\sqrt{3}$ B) $\sqrt{3}$ C) $3\sqrt{3}$ D) $2\sqrt{3}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme: Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. A bag contains 40 white balls and 20 red balls. 30 balls are drawn one by one randomly from the bag with replacement. If X be the number of white balls drawn, then $\left(\frac{\text{mean of X}}{\text{standard deviation of X}}\right)$ is $\sqrt{k}$, then k is _____

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72. If a function satisfy the equation $\cos^2 x\frac{dy}{dx}-(\tan 2x)y=\cos^4 x$; passing through $\left(\frac{\pi}{6},\frac{3\sqrt{3}}{8}\right)$ then the value of $64y^2\left(\frac{\pi}{3}\right)$ is _____.

73. Let $f(x)$ be a monic quadratic polynomial with $f(0)=0$ and $f(1)=1$. Then $\lim\limits_{x\to 0}\frac{f(x)(2^{\sin x}+3^{\tan x}-2)}{x-\sin x}=k\ln k$, then $k=$ _____

74. The maximum value of $(\sqrt{-3+4x-x^2}+4)^2+(x-5)^2$ is _____ [Where $1\le x\le 3$]

75. If the equation $|2-x|-|x+1|=k$ has exactly one solution, then the number of integral values of k is ___

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